Introduction to programming

Lab05

Midterm test

- 27-31of October 2025
  - During the labs
  - Paper based (30 minutes)
    - Number systems, conversions (4 task)
    - Decoding/Explaining codes (4 task)
  - Computer based (60 minutes)
    - Python program (4 task)
    - You are not allowed to use your own laptop!

Revision

- Input numbers/Reading values
  - Until 0
  - Until a negative number
  - Until EOF -> End-Of-File (CTRL+D)
- Exceptions
  - try – except
  - try – finally
- Strings
  - String functions
  - Slicing strings

Homework 1

- Write a program that determines and prints how many **positive even values** were present in the input.
- The first line contains a positive integer n, representing how many numbers follow.
  Each of the next n lines contains one integer.
- The program should write exactly one integer to the output, the count of positive even numbers.
- **Input:**
  - 5
  - -2
  - -1
  - 0
  - 1
  - 2
- **Output:**
  - 1

Solution

```
n = int(input())
c = 0

for i in range(n):
    num = int(input())
    if num > 0 and num % 2 == 0:
        c += 1

print(c)
```

Homework 2

- Write a program that reads words from the input until *EOF* and removes all **vowels** from each word.
- `Input:`
  - `apple`
  - `pear`
  - `peach`
- `Output:`
  - `ppl`
  - `pr`
  - `pch`

Solution – 1

```
import sys

for s in sys.stdin:
    r = ''
    for c in s.strip():
        if c.lower() not in "aeiou":
            r = r + c
    print(r)
```

Solution – 2

```
import sys

for s in sys.stdin:
    r=''.join([c for c in s.strip() if c.lower() not in "aeiou"])
    print(r)
```

Solution – 3

```
while True:
    try:
        s = input()
        r = ''
        for c in s:
            if c.lower() not in "aeiou":
                r = r + c
        print(r)
    except EOFError:
        break
```

Solution – 4

```
while True:
    try:
        s = input()
        r=''.join([c for c in s if c.lower() not in "aeiou"])
        print(r)
    except EOFError:
        break
```

Solution – 5

```
import sys

for s in sys.stdin:
    s = s.strip()
    for c in s:
        if c.lower() in 'aeiou':
            s = s.replace(c, '')
    print(s)
```

Solution – 6

```
import sys

for s in sys.stdin:
    s = s.strip()
    i = 0
    while i < len(s):
        if s[i].lower() in 'aeiou':
            s = s.replace(s[i], '')
        i = i + 1
    print(s)
```

Homework 3

- Write a program that reads until the string "END" is entered, and duplicates all uppercase letters in each word.

- Input:
  - Apple
  - PeaR
  - PeaCh
  - END

- Output:
  - AApple
  - PPeaRR
  - PPeaCCh

Solution – 1

```
s = input()
while s != 'END':
    r = ''
    for c in s:
        if c.isupper():
            r += c * 2
        else:
            r += c
    print(r)
    s = input()
```

Solution – 2

```
while True:
    s = input()
    if s == 'END':
        break
    r = ''
    for c in s:
        if c.isupper():
            r += c * 2
        else:
            r += c
    print(r)
```

Solution – 3

```
s = input()
while s != 'END':
    r = ''.join([c * 2 if c.isupper() else c for c in s])
    print(r)
    s = input()
```

Exercise – GCD
https://progcont.hu/progcont/b50/?pid=200311

- Write a program that reads two positive integers from standard input and determines their greatest common divisor (**GCD**).

Input Specification

- The input contains two positive integers separated by a space character.

Output Specification

- The program should write a single line to the standard output containing the greatest common divisor of the two input number.

Sample Input

- 15 10

Output for Sample Input

- 5

GCD – Euclidean Algorithm

- The Euclidean algorithm is based on the principle that the greatest common divisor of two numbers does not change if the **larger number is replaced by its difference with the smaller number.**
- For example, 21 is the GCD of 252 and 105 (as $252 = 21 \times 12$ and $105 = 21 \times 5$), and the same number 21 is also the GCD of 105 and $252 - 105 = 147$.
- Since this replacement reduces the larger of the two numbers, repeating this process gives successively smaller pairs of numbers until the **two numbers become equal.**

a	b
252	105
147	105
42	105
42	63
42	21
21	**21**

Solution 1

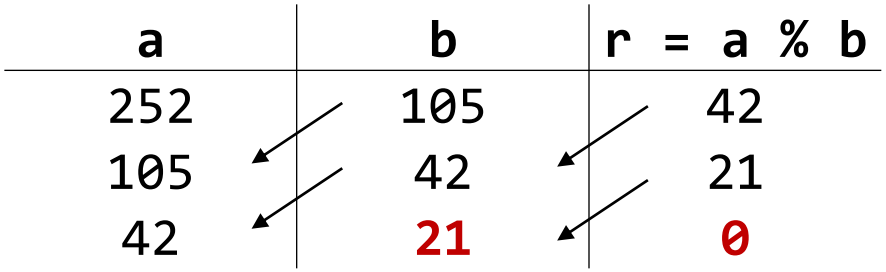
```
line = input()
s = line.split(" ")
a = int(s[0])
b = int(s[1])

while a != b:
    if a > b:
        a = a - b
    else:
        b = b - a

print(b)
```

GCD – Euclidean Algorithm

- A more efficient version of the algorithm shortcuts these steps, instead replacing the larger of the two numbers by its remainder when divided by the smaller of the two (with this version, the algorithm stops when the remainder becomes zero).



Solution 2

```
line = input()
s = line.split(" ")
a = int(s[0])
b = int(s[1])

while a % b != 0:
    r = a % b
    a = b
    b = r

print(b)
```

Solution 2

```
line = input()
s = line.split(" ")
a = int(s[0])
b = int(s[1])

while a % b:
    r = a % b
    a = b
    b = r

print(b)
```

Solution 3

```
line = input()
s = line.split(" ")
a = int(s[0])
b = int(s[1])
gcd = 1
if a < b:
    c = a
else:
    c = b
for i in range(2, c + 1):
    if (a % i == 0 and b % i == 0):
        gcd = i

print(gcd)
```

Least Common Multiple

- The least common multiple, lowest common multiple, or smallest common multiple of two integers a and b, is the smallest positive integer that is divisible by both a and b.

- $$LCM = \frac{a \cdot b}{GCD}$$

Exercise – Relative Primes
https://progcont.hu/progcont/b50/?pid=200312

- Write a program that reads two positive integers from the standard input and decides whether they are relative primes or not.

Input Specification

- The input contains two positive integers separated by a space character.

Output Specification

- The program should write a single line to the standard output containing the string „IGEN” („yes” in Hungarian) or „NEM” („NO” in Hungarian) depending on whether the two input numbers are relative primes.
- **Sample Input**
- 15 10

Output for Sample Input

- NEM

Solution

```
line = input()
s = line.split(" ")
a = int(s[0])
b = int(s[1])
while a != b:
    if a > b:
        a = a - b
    else:
        b = b - a

if a == 1:
    print("IGEN") #yes
else:
    print("NEM") #no
```

Random numbers

- we often use random numbers in mathematical and statistical problems
- Python includes a random number generator
- `import random` – import module
- Random module functions:
  - `random()` – returns a random **float** between [0.0, 1.0)
  - `uniform(a,b)` – returns a random **floating number** between the two specified numbers (both included) [a,b]
  - `randint(a,b)` – returns a random **integer** number between the two specified numbers (both included) [a,b]
  - `choice()` – returns a randomly selected element from the specified sequence.

Exercise

- Write a program that calculates the product of n randomly generated integers between [1,10].

Solution

```
import random

n = int(input("n = "))
p = 1
for i in range(n):
    x = random.randint(1, 10)
    print(x, end=' ')
    p = p * x
print("\nProduct = ", p)
```

Exercise

- Write a program that continuously generates random integers between 1 and 10, and stops when the number **5** appears.
- Print how many steps it took.

Solution

```
import random

c = 0
while True:
    x = random.randint(1, 10)
    if x == 5:
        break
    print(x, end=' ')
    c += 1
print("\nStopped after {} steps!".format(c))
```

Lists

- A list is a collection of values, in fact a container that can contain numbers, texts, etc.
- The values that make up a list are called elements.
- The elements are numbered, **indexed**.
- The elements of a list can be of any type.
- Create an empty list:

```
list = []
```

Lists

- Length of the list: `len()`
- List traversal: `a = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]`

```
for i in a:
    print(i, end=' ')

for i in range(len(a)):
    print(a[i], end=' ')

i = 0
while i < len(a):
    print(a[i], end=' ')
    i += 1
```

List operations

- + operator concatenates lists
- * operator repeats a list a specified number of times
- slicing operator: remove items from a list by assigning an empty list to a slice of the list or add an item to a list by inserting an empty slice at the desired location.

Lists

```
>>> [1,2,3]
[1, 2, 3]
>>> a = [1,2,3]
>>> list = []      #empty list
>>> list
[]
>>> a = [1,2,'ab',3.14]
>>> a
[1, 2, 'ab', 3.14]
>>> len(a)         #length of the list
4
>>> [1,2]+[3,4]    #concatenate
[1, 2, 3, 4]
```

Lists

```
>>> a=[1,2,3]
>>> b = a        # refer to a reference
>>> b
[1, 2, 3]
>>> a[0]
1
>>> a[0]=10
>>> a
[10, 2, 3]
>>> b
[10, 2, 3]
>>> b=a[:]       # a full copy of a
>>> b
[10, 2, 3]
>>> a[0]=20
>>> b
[10, 2, 3]
>>> a
[20, 2, 3]
```

Slicing lists

```
>>> a
[20, 2, 3]
>>> a == b
False
>>> a[1:]
[2, 3]
>>> a[-1]
3
>>> a[:1]
[20]
>>> a*3
[20, 2, 3, 20, 2, 3, 20, 2, 3]
```

List operations

```
>>> a=[1,2,3]
>>> a.append(20) #adds a new element to the end
>>> a
[1, 2, 3, 20]
>>> a.pop(0) #removes the element at index 0, returns
its value
1
>>> a
[2, 3, 20]
>>> del a[2]
>>> a
[2, 3]
```

List operations

```
>>> a=[1,2,3,4,5,6,7,8]
>>> a[2:4]
[3, 4]
>>> a[2:4]=[] #remove elements
>>> a
[1, 2, 5, 6, 7, 8]
>>> a[2:4]=[3,1,2] #insert elements
>>> a
[1, 2, 3, 1, 2, 7, 8]
```

Sorting the list

```
>>> a=[5, 2, 4, 8, 3, 1]
>>> a
[5, 2, 4, 8, 3, 1]
>>> sorted(a)
[1, 2, 3, 4, 5, 8]
>>> sorted(a, reverse=True)
[8, 5, 4, 3, 2, 1]
>>> a
[5, 2, 4, 8, 3, 1]
>>> a = sorted(a)
>>> a = ['c','b','a','d']
>>> a.sort()
>>> a
['a', 'b', 'c', 'd']
```

List operations

```
>>> list = [9, 3, 4, 6, 8, 5, 3, 1, 5]
>>> max(list)
9
>>> min(list)
1
>>> sum(list)
44
>>> list.count(3)
2
```

List operations

- `append(element)` – appends the element specified as argument to the end of the list. The list must already exist, e.g. an empty list.
- `insert(position, element)` – inserts the element given as argument into the list at the given position, in which case the rest of the list moves one position to the right of the given position.
- `extend(list)` – appends all elements of the list specified as argument to the end of the list.
- `clear()` – clears all elements of the list (does not delete the list itself).

List operations

- `remove(element)` – deletes the first occurrence of the item specified as argument from the list. If the argument does not occur in the list, the program terminates with a runtime error.
- `pop(index)` – removes and returns the element at the given index.
- `del list[index]` – deletes the element at the given index.
- `count(element)` – Counts how many times the element (specified as argument) appears in the list.

List operations

- `index(element)` returns the index of the first occurrence of the element in the list. If the element given as argument is not in the list, it raises an error message.
- `reverse()` – reverses the elements in the list.
- `sort()` – sorts the elements in the list.
  If the list elements are numbers, they are in ascending order, if they are strings, they are in alphanumeric order.
  If the elements of the list are of mixed type, it terminates with a runtime error.

List Comprehension

The Syntax:

- newlist =
  [*expression* for *item* in *iterable* if *condition* == True]

```
>>> list = [x for x in range(10)]
>>> list
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]

>>> newlist = [x for x in list if x % 2 == 0]
>>> newlist
[0, 2, 4, 6, 8]
```

List Comprehension

```
fruits =
["apple", "banana", "cherry", "kiwi", "mango"]
newlist = []
for x in fruits:
  if "a" in x:
    newlist.append(x)

#newlist = [x for x in fruits if 'a' in x]

print(newlist)
```

Examples

```
fruits = ["apple", "banana", "cherry",
"kiwi", "mango"]

newlist = [x.upper() for x in fruits]

print(newlist)
```

Exercise – Practice

- The following tasks should be solved using "list comprehension".

Task1

- ['anthony', 'blake', 'charlotte'] → ['Anthony', 'Blake', 'Charlotte']

Task2

- [0, 0, 0, 0, 0, 0, 0, 0, 0, 0], i.e. initialize a list of 10 zeros.

Task3

- [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] → [2, 4, 6, 8, 10, 12, 14, 16, 18, 20]

Task4

- ['1', '2', '3', '4', '5', '6', '7', '8', '9', '10'] → [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] (first list contains strings)

Exercise – Practice

- The following tasks should be solved using "list comprehension".

Task5

- "1234567" → [1, 2, 3, 4, 5, 6, 7] i.e. we have numbers in string format, and we want to put the digits (as numbers) in a list

Task6

- 'The quick brown fox jumps over the lazy dog' → [3, 5, 5, 3, 5, 4, 3, 4, 3], i.e. the length of each word

Task7

- "python is an awesome language" → ['p', 'i', 'a', 'a', 'l'], i.e. collect the initial letters of each word in a list

Solutions

Task1

- ['anthony', 'blake', 'charlotte'] → ['Anthony', 'Blake', 'Charlotte']

```
#Solution 1
l = ['anthony', 'blake', 'charlotte']
new = [x[0].upper() + x[1:] for x in l]
print(new)

#Solution 2
l = ['anthony', 'blake', 'charlotte']
new = [x.capitalize() for x in l]
print(new)
```

Solutions

Task2

- [0, 0, 0, 0, 0, 0, 0, 0, 0, 0], i.e. initialize a list of 10 zeros.

```
new = [0 for i in range(10)]
print(new)
```

Task3

- [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] → [2, 4, 6, 8, 10, 12, 14, 16, 18, 20]

```
l = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
new = [2 * i for i in l]
print(new)
```

Solutions

Task4

- ['1', '2', '3', '4', '5', '6', '7', '8', '9', '10'] → [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] (first list contains strings)

```
l = ['1', '2', '3', '4', '5', '6', '7', '8', '9', '10']
new = [int(i) for i in l]
print(new)
```

Task5

- "1234567" → [1, 2, 3, 4, 5, 6, 7] i.e. we have numbers in string format and we want to put the digits (as numbers) in a list

```
s = "1234567"
new = [int(c) for c in s]
print(new)
```

Solutions

Task6

- 'The quick brown fox jumps over the lazy dog' → [3, 5, 5, 3, 5, 4, 3, 4, 3], i.e. find the length of each word

```
s = 'The quick brown fox jumps over the lazy dog'
new = [len(c) for c in s.split()]
print(new)
```

Task7

- "python is an awesome language" → ['p', 'i', 'a', 'a', 'l'], i.e. collect the initial letters of each word in a list

```
s = 'python is an awesome language'
new = [c[0] for c in s.split()]
print(new)
```

Exercise

- Fill a list with n **even numbers** randomly generated between 1 and 10.

Solution

```
import random

n = int(input("n="))
list = []
while True:
    if len(list) == n:
        break
    x = random.randint(1, 10)
    if x % 2 == 0:
        list.append(x)
print(list)
```

Exercise

- Create a list of n random integers between 1 and 10, then:
  - print the list
  - print the sum of the elements
  - print the minimum and the maximum of the elements
  - check whether all elements are even
  - sort the even elements in ascending order
  - sort the odd elements in descending order

```python
import random

list = []
evens = True
even = []
odd = []
n = int(input('n='))
for i in range(n):
    x = random.randint(1, 10)
    list.append(x)
    if x % 2 != 0:
        odd.append(x)
        evens = False
    else:
        even.append(x)

print(list)
print("Sum =", sum(list))
print("Evens =", evens)
print("Min =", min(list))
print("Max =", max(list))
print("Even elements =", sorted(even))
print("Odd elements =", sorted(odd, reverse=True))
```

```
import random

n = int(input('n='))
list = [random.randint(1, 10) for i in range(n)]
even = [i for i in list if i % 2 == 0]
odd = [i for i in list if i % 2 != 0]
evens=False
if even == list:
    evens = True

print(list)
print("Sum =", sum(list))
print("Evens =", evens)
print("Min =", min(list))
print("Max =", max(list))
print("Even elements =", sorted(even))
print("Odd elements =", sorted(odd, reverse=True))
```

Homework 1 – Neanderthal gene
https://progcont.hu/progcont/100350/?pid=201515

Neanderthal gene

- A recent study suggests that the risk of coronavirus infection is more severe in people who carry a small Neanderthal gene. Reading this makes you stop for a moment and wonder if you have even an ounce of the Neanderthal gene in you.
- If you make a model, Neanderthal genes are relatively easy to identify. Suppose we describe a DNA molecule by a sequence of integers. If we can find at least three consecutive values in this sequence that form a strictly monotonic series, we can say that the DNA molecule contains a Neanderthal gene.
- Write a program that reads at least one positive integer per line from the standard input to the end of file (EOF). Each line describes a DNA molecule. If the DNA molecule under test contains a Neanderthal gene, your program should write a line containing a "YES" string to the standard output, otherwise a "NO" string.

Homework 1 – Neanderthal gene

https://progcont.hu/progcont/100350/?pid=201515

Sample Input

```
5 6 7
9 8 7 6 5
2 3
7 11 8 10 9
4 4 4 4
9
1 2 3 1 2
7 8 7 8 7 8 9
```

Output for Sample Input

```
YES
YES
NO
NO
NO
NO
YES
YES
```

Homework 2 – Piggy bank
https://progcont.hu/progcont/100353/?pid=201527

- **Piggy bank**
- Children love to collect their savings in piggy banks. In this exercise, you have to make a statement of these savings.
- Write a program that reads the names of the children and the money they have saved from week to week from standard input until end-of-file (EOF), line by line!
- The input format is:
- `name:quantity[ quantity]...`
- Your program should write in the **first line** of the standard output the total amount of money saved by all children! While processing the data, also count the children who regularly saved at least 20 forints in their piggy bank every week! On the **second line**, print the number of children who saved **at least 20 forints** every week.

Homework 2 – Piggy bank

https://progcont.hu/progcont/100353/?pid=201527

Sample Input

- Mark:20 30 40 50 60
- Jogh:25 15 25 15 25 15
- Peter:30
- David:20 30 10 30 20 10 20 30
- Julie:19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

Output for Sample Input

- 710
- 2

Homework 3 – Reverse Numbers
https://progcont.hu/progcont/100348/?pid=201508

- **Reverse numbers**
- Write a program that reads a sequence of numbers from the standard input and writes its elements in reverse order to the standard output.
- Each line of the input contains at least one integer, and the consecutive numbers are separated by exactly one space.
  The first number (m) in each line is the number of elements in the number line to be processed.
  The program must write the following sequence of m integers in reverse order to the standard output. The end of the input is indicated by a line containing only one 0.
  he program need not process this line.
- Make sure that the numbers written in a line are separated by exactly one space, and do not print spaces at the beginning and end of the line.

Homework 3 – Reverse Numbers

https://progcont.hu/progcont/100348/?pid=201508

Sample Input

- 4 1 2 3 4
- 5 2 5 3 4 1
- 2 2 3
- 0

Output for Sample Input

- 4 3 2 1
- 1 4 3 5 2
- 3 2